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CONSORT RANDOMIZED CLINICAL TRIAL

Clinical and Radiographic Assessments of Potassium Nitrate in Polycarboxylate Versus Mineral Trioxide Aggregate as Pulpotomy Biomaterials in Immature Mandibular First Permanent Molars: A Randomized Clinical Trial

SIGNIFICANCE

Pulpotomy in asymptomatic vital exposed immature mandibular first molars using potassium nitrate in polycarboxylate cement and mineral trioxide aggregate achieves similar and promising results within the 12-month follow-up.

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KEY WORDS

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There are great challenges in the management of permanent teeth with incomplete root development having exposed vital compromised coronal pulp. These cases can be treated with vital pulp therapy, namely direct pulp capping and pulpotomy¹. In the latter approach, the coronal pulp is amputated, and the radicular tissues are treated with a biomaterial to allow complete apical root development². A systematic review¹ revealed that full pulpotomy in permanent teeth with cariously exposed pulp would have a high success rate over 3 years with fluctuation of the success rate occurring in direct pulp capping cases. An important benefit for the preservation of the radicular vital pulp is the protective resistance to mastication forces and fracture compared with a root canal-filled tooth³.

Different pulpotomy materials have been used in practice, with the most popular being mineral trioxide aggregate (MTA)⁴. Clinical studies evaluated the use of MTA as a direct pulp capping material⁵ and as a pulpotomy material after partial and complete pulpotomy^{6–9}. Despite its clinical effectiveness, MTA has some drawbacks such as difficult handling characteristics¹⁰, a long setting time¹¹, tooth discoloration¹², and high cost. These urged the search for an alternative pulpotomy material to MTA.

Potassium nitrate (KNO₃) is a known desensitizer for hypersensitive teeth¹³. Histopathologic animal studies reported a satisfactory response for the use of polycarboxylate as a direct pulp capping material^{14,15}. The combination of 5% KNO₃ in polycarboxylate cement is reported for direct capping of traumatically exposed pulp in animal models¹⁶.

Clinical studies evaluated KNO₃ in polycarboxylate in vital pulp therapy as a pulp capping^{17–19} or pulpotomy material²⁰. Clinical and radiographic evaluation of teeth treated with KNO₃ in polycarboxylate cement as a direct pulp capping material for deep nonexposed carious lesions showed a success rate of 98% over a 4-year period¹⁷. Direct capping of KNO₃/dimethyl isosorbide/polycarboxylate cement in carious exposure vital pulp showed all patients to be clinically and radiographically normal over 2 years¹⁸. Indirect capping by 5% KNO₃ in polycarboxylate cement was compared with calcium hydroxide liner in the treatment of reversible pulpitis in permanent teeth. The results suggested that 5% KNO₃ in polycarboxylate cement had a very good analgesic action with normal electrical excitability being restored faster than cases treated with calcium hydroxide¹⁹. A 2-step

technique of pulpotomy in permanent teeth used ferric sulfate followed by a mixture of either iodoform or KNO₃. After a 2-year observation period, the author concluded that both formulations had a very promising value with the production of a dentin bridge²⁰.

KNO₃ in polycarboxylate has the added benefit of low cost and easy material handling. However, no previous randomized clinical trials (RCTs) have evaluated the use of polycarboxylate containing KNO₃ as a pulpotomy material. Thus, the trial aimed to set an evidence-based decision by comparing in an RCT the clinical/radiographic outcomes of KNO₃ in polycarboxylate cement and MTA as pulpotomy biomaterials used for asymptomatic vital immature permanent mandibular first molars.

METHODS

The study design was a parallel, randomized 1:1 allocation ratio RCT and was designed in accordance with the Consolidated Standards of Reporting Trials guidelines. The participants, their legal guardians, and the data assessor were blind to the intervention used. The protocol was approved by the Department of Endodontics, Evidence-base Committee and Ethical Committee, Faculty of Dentistry, Cairo University, Cairo, Egypt. The study protocol was registered on www.clinicaltrials.gov (NCT03166748).

Guided by Nosrat et al⁶ and Keswani et al⁷, a power analysis was conducted, and the sample size was set at 50 participants (25 in each group) based on the probability of a type I error of 0.05 and power at 0.8. If there is truly no difference between the control and experimental treatment, then 42 patients are required to be 80% sure that the limits of a 2-sided 95% confidence interval will exclude a difference between the standard and experimental biomaterial groups of more than 10%. This number was increased to 50 to compensate for losses during follow-up.

Participants were recruited from the outpatient clinic of the Department of Pedodontics, Faculty of Dentistry, Cairo University. The eligibility criteria were children ranging from 6–9 years who had an asymptomatic vital immature mandibular first molar with extensive and deep caries that revealed pulp exposure during caries excavation. Cases had no preoperative acute pulp or periapical involvement symptoms. The exclusion criteria were medically compromised children and cases diagnosed to have necrotic pulp or acute inflammation of the pulp or the periapical tissue. Additionally, cases having abnormal tooth anatomy, furcation

involvement, canal calcification, or a nonrestorable tooth were excluded. The nature of the treatment and the risks were explained in detail to the guardians of the children. Written consent was obtained from the guardians before the treatment.

A detailed history of participants was taken followed by a thorough clinical and radiographic examination. Visual pulp status examination revealed the involved tooth to have extensive caries. The tooth was initially assessed by thermal stimulation. A cold test was performed using Endo-Ice spray (Maquira, Maringá, Brazil), which was done by applying cold-sprayed cotton on the tested tooth compared with the contralateral tooth. Thereafter, if excavation ended up with exposure, access cavity preparation was reported. Teeth with excessive uncontrolled bleeding or the absence of bleeding during the access cavity were excluded.

Preoperative radiographic assessments checked the extent and location of caries as well as the periapical status. Radiographs were taken by an x-ray machine (Vivi SRL, Tolentino, Italy) using the Digital ScaNeo imaging plate size 2 (FONA, Bratislava, Slovakia). A standardized paralleling technique was performed with the Rinn XCP alignment system (Rinn Corporation, Elgin, IL).

A total of 50 molars from 48 children were randomly allocated into 2 groups, each composed of 25 molars. The random sequence was generated by the main supervisor (<http://www.random.org/>). Patients were anesthetized by an inferior alveolar nerve block using a single cartridge (1.8 mL) containing mepivacaine hydrochloride and levonordefrin hydrochloride. The tooth was isolated with a rubber dam. Caries were excavated by a spoon excavator until pulp exposure was evident. Access cavity preparation was performed with a sterile high-speed carbide round bur and an Endo-Z bur (Maillefer, Dentsply, Switzerland). The pulp was amputated with a sterile #5 round bur and a sharp spoon excavator up to the canal orifice. To achieve complete hemostasis, a sterile gauze soaked in saline was initially placed on the amputated pulp for 5–10 minutes followed by the placement of ferric sulfate 15.5% gel for 15 seconds²⁰. Ferric sulfate 15.5% gel is a hemostatic agent that is provided in a plastic syringe and a cotton-tipped disposable needle (Viscostat; Ultradent Products Inc, Salt Lake City, UT). A small amount of ferric sulfate was applied by gently wiping the disposable cotton tip of the needle against the amputated pulp for 15 seconds. It was flushed from the pulp chamber with saline in order to remove any remnants of blood clot formation. The

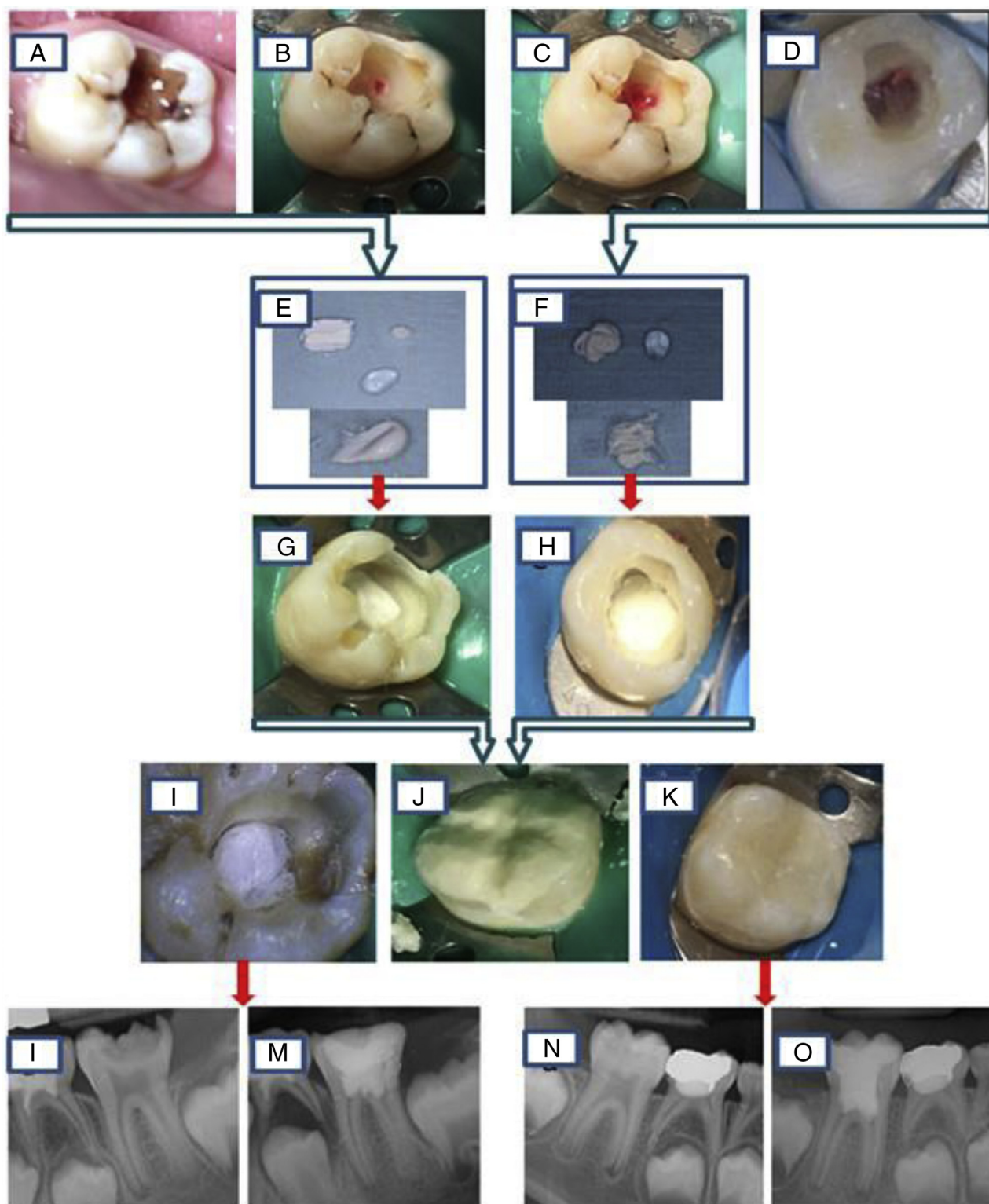


FIGURE 1 – Procedural steps for the application of KNO_3 /polycarboxylate (the intervention group) and MTA (the control group). (A) A carious mandibular first molar. (B) Pulp exposure after caries excavation. (C) Access cavity preparation. (D) Control of bleeding with ferric sulfate. (E) Mixing of KNO_3 /poly carboxylate. (F) Mixing of MTA. (G) Placement of KNO_3 /polycarboxylate in the pulp chamber. (H) Placement of MTA. (I) Cotton pellet application. (J) Temporization by glass ionomer. (K) Final composite restoration. (L) A preoperative radiograph for a case in the intervention group. (M) A postoperative radiograph after KNO_3 /poly carboxylate placement and final restoration. (N) A preoperative radiograph for a case in the control group. (O) A postoperative radiograph after MTA placement and final restoration.

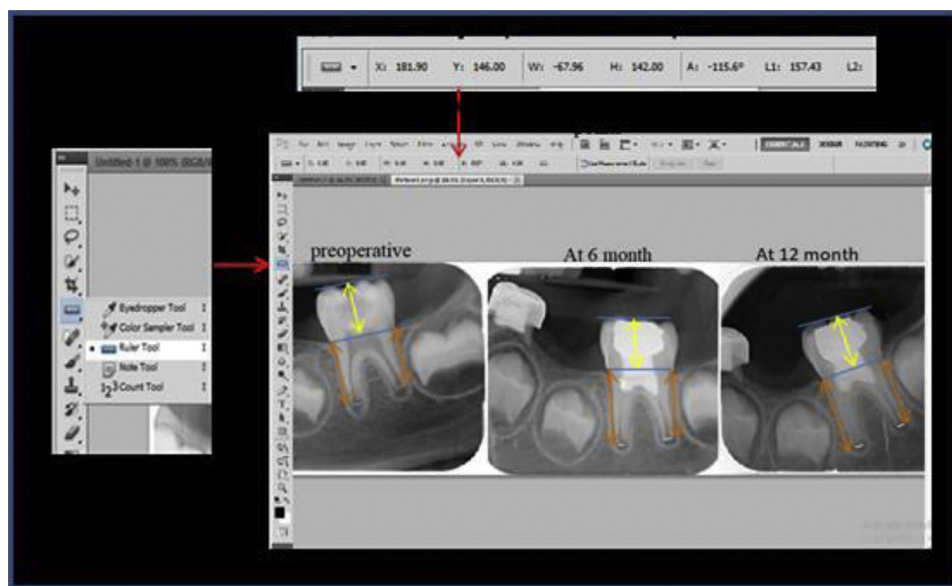


FIGURE 2 – Procedural steps for obtaining measurements for the root length and apical foramen diameter; the *yellow lines* represent the crown height, the *orange lines* represent the root length, and the *white lines* represent the foramen width. For both the mesial and distal roots, the following constructions and measurements were done in preoperative and 6- and 12-month radiographs. (A) Measuring the root length: 2 equal horizontal lines were constructed, 1 at the level of the cervical line and another one at the level of the radiographic root apex. The perpendicular distance measured between both lines is the root length. (B) Measuring the apical foramen diameter: the horizontal line was constructed between the canal walls at the open apex level of the mesial and distal roots and measured. (C) Checking and adjusting the magnification discrepancy: this was done by comparing crown heights within the 6- and 12-month radiographs relative to the preoperative one. Two equal horizontal lines were constructed, one at the level of the cervical line and another one at the occlusal surface. The perpendicular distance measured between both lines is the crown height. (D) Calculating the magnification factor: the postoperative crown lengths at 6 and 12 months were subtracted from the preoperative value and divided by preoperative value. Magnification discrepancy of the postoperative measurements was adjusted by multiplying with the previously calculated magnification factor. (E) Measurements obtained in pixels were converted into millimeters; initially, the numbers of pixels for a known length of a metal ruler (10 mm) were recorded (223 pixels). Thus, any measurement in pixels was simply multiplied by 10 and divided by 223 to convert it to a millimeter value.

randomization sequence was kept in 8 folded paper in a sealed envelope. The operator opened the envelope after hemostasis at the pulpotomy agent placement step. The materials were as follows: in the intervention group, teeth were treated with 5% KNO₃ (Merck, Darmstadt, Germany) and polycarboxylate cement (SIMENT; Deepak Product, Inc, Medley, FL) pulpotomy, and in the control group, teeth were treated with MTA pulpotomy (white MTA; Angelus, Londrina, PR, Brazil).

In the intervention group, the cavity was covered with 5% KNO₃ in polycarboxylate. A pilot study showed that a mix of 100 mg powder (zinc oxide and KNO₃) with the liquid polyacrylic acid was sufficient to achieve adequate thickness of the pulpotomy material. To prepare 5% KNO₃ in polycarboxylate, preweighed 95 mg zinc oxide powder and 5 mg KNO₃ were placed on a glass slab followed by mixing with polyacrylic acid (liquid) in a 1:1 powder to liquid ratio. A puttylike consistency was obtained with a working time of approximately 1 minute 45 seconds. The mixed material was placed over pulp tissue in a thickness of about 3–4 mm using a Teflon-coated hand instrument, and light pressure was applied with a wet cotton pellet. The cavity

was temporized with glass ionomer filling material (Medifil; Promedica, Neumünster, Germany).

In the control group, the cavity was covered with white MTA. The mixed material (according to the manufacturer's instructions) was placed over pulp tissue in a thickness of about 3–4 mm in the access cavity using an MTA applicator (Angelus, Londrina, Brazil) and condensed lightly with a moistened sterile cotton pellet. Because the setting time of MTA is about 4 hours, the MTA mixture was covered with a moistened cotton pellet, and the cavity was temporized with glass ionomer filling material (Medifil).

A periapical radiographic image was taken to ensure the level of the material at the canal orifices. The following day the patient was recalled, and glass ionomer filling and the cotton pellet were removed in the control group. A resin-modified glass ionomer (Prime Dental Manufacturing, Chicago, IL) was placed as a base to seal the pulp chamber. For both groups, the cavity was restored with resin composite (Filtek Z350 XT Universal Restorative; 3M ESPE, St Paul, MN) as a permanent restoration. Figure 1A–O shows the procedural steps for material application in both groups.

The primary outcomes were clinical and radiographic outcomes assessed by a direct answer of yes or no with no subjective influence. Clinical evaluations checked clinical signs and symptoms of spontaneous pain, sensitivity to percussion/palpation, and soft tissue swelling or a sinus tract by binary answers (no [success] or yes [failure]) at 3 follow-up periods (1 week [by phone] and 6 and 12 months). Radiographic evaluations checked for root maturation progress denoting success as well as checked signs of failure (including the formation of a periradicular/interradicular lesion and internal/external root resorption) by binary answers yes or no at 2 follow-up periods (ie, 6 and 12 months). In case of a postoperative emergency (pain and/or swelling), an emergency appointment was scheduled, treatment was shifted to total root canal treatment, and the case was excluded from the study. The main researcher knew the pulpotomy agent because every material has its own handling properties; he also performed the treatment as well as the clinical and radiographic assessments.

For the secondary outcome, radiometric analysis was performed by the operator for preoperative, 6-month, and 12-month

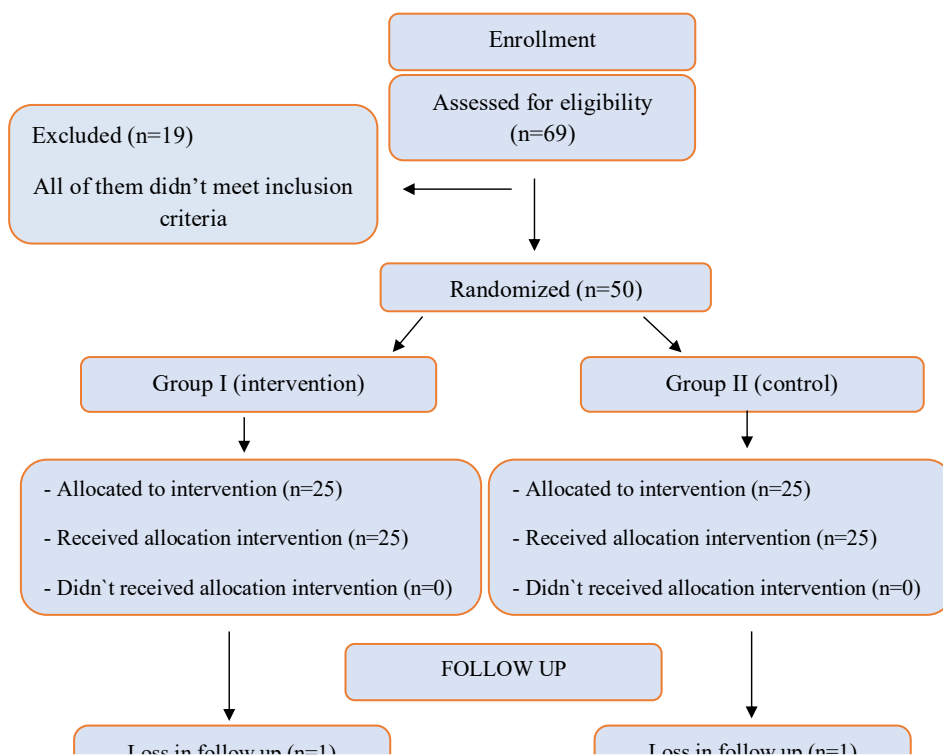
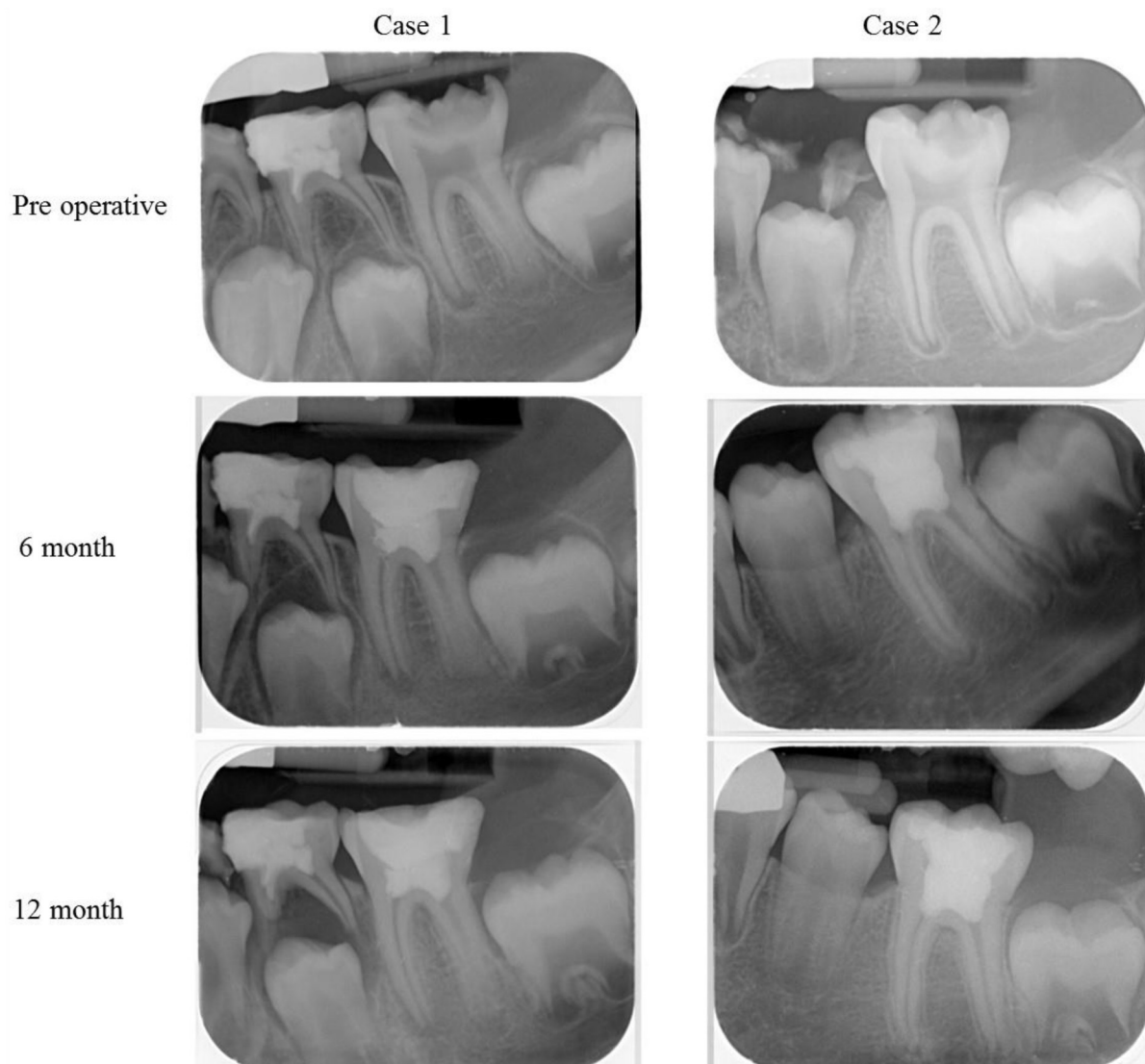


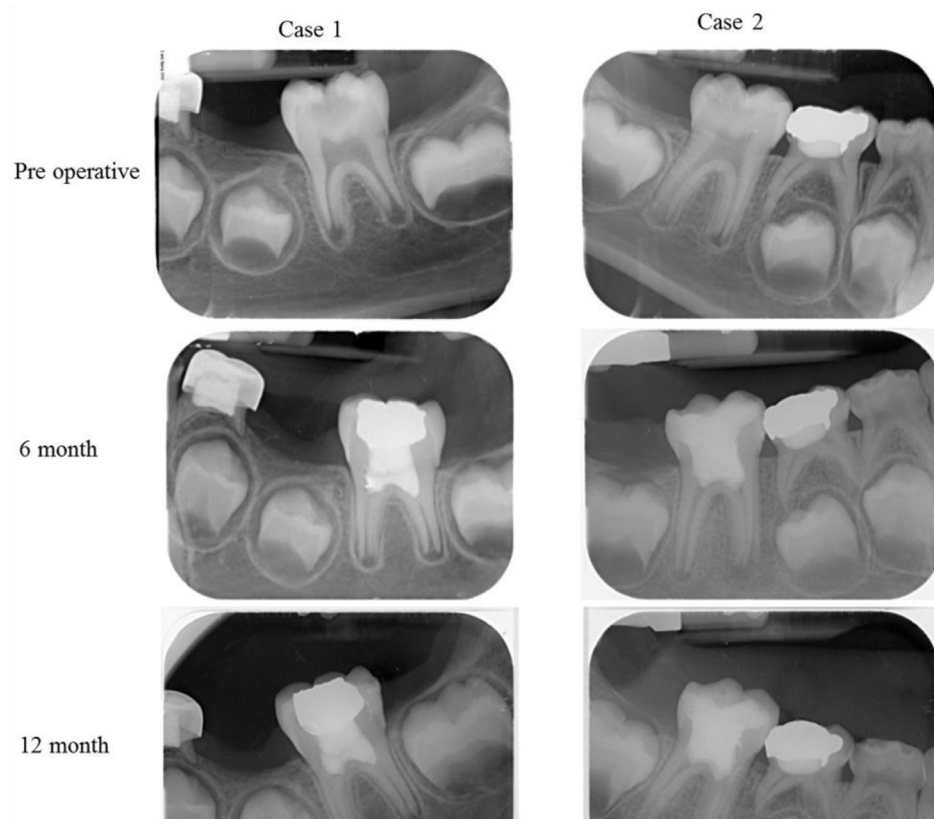
FIGURE 3 – Consolidated Standards of Reporting Trials flow diagram of the study.

radiographs; the radiographs were imported and measured for data analysis using Photoshop software (Photoshop CS5 Extended ME; Adobe, San Jose, CA). Blinding during radiographic assessment was not feasible because the 2 pulpotomy materials could be distinguished radiographically.

Figure 2 shows the procedural steps for measuring the root length and the apical foramen diameter. The percentage difference between the preoperative and postoperative radiographs was calculated for both the mesial and distal roots at 6 months and 12 months. At 12 months, roots showing complete apical closure were counted and compared with roots showing root development yet to be achieved.

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